

DTSA 5001

Probability Foundations for Data Science and AI

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – Statistics Pathway	ML Engineer (39.9 / 100)	31.8 / 100
SPECIALIZATION Data Science Foundations: Statistical Inference Pathway		

32

CURRENT AVG SCORE
across 6 job roles

65

POTENTIAL SCORE
with recommended additions

+44

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Probability Theory covers the foundations of probability and its relationship to statistics and data science. Calculate a probability, independent and dependent outcomes, and conditional events. Understand discrete and continuous random variables and see how this fits with data collection. Learn Gaussian (normal) random variables and the Central Limit Theorem and understand it’s fundamental importance for statistics and data science.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE		PROJECTED SCORE		POINT GAIN	PRIORITY
ML Engineer		39.9		68.0	+28.2	MEDIUM
AI Engineer		35.7		70.6	+34.9	HIGH
Data Scientist		34.2		57.2	+23.0	MEDIUM
Machine Learning Engineer		33.5		68.8	+35.3	HIGH
Applied Scientist		28.2		72.5	+44.3	HIGH
MLOps Engineer		19.3		52.4	+33.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

ML Engineer	39.9 → 68.0 (+28.2 pts)
learning machine systems problems new technical building tools	
AI Engineer	35.7 → 70.6 (+34.9 pts)
following learning software technical systems teams problems machine	
Data Scientist	34.2 → 57.2 (+23.0 pts)
sets presentation problems	
Machine Learning Engineer	33.5 → 68.8 (+35.3 pts)
sets learning tools exploratory development applications machine technical	
Applied Scientist	28.2 → 72.5 (+44.3 pts)
computational research interdisciplinary problems technical department learning systems	
MLOps Engineer	19.3 → 52.4 (+33.0 pts)
systems learning machine tools development error	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.